Fundamentals of AI and ML
(CSA2001)

Project Report: AI Travel Planner

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Year: 1st Year

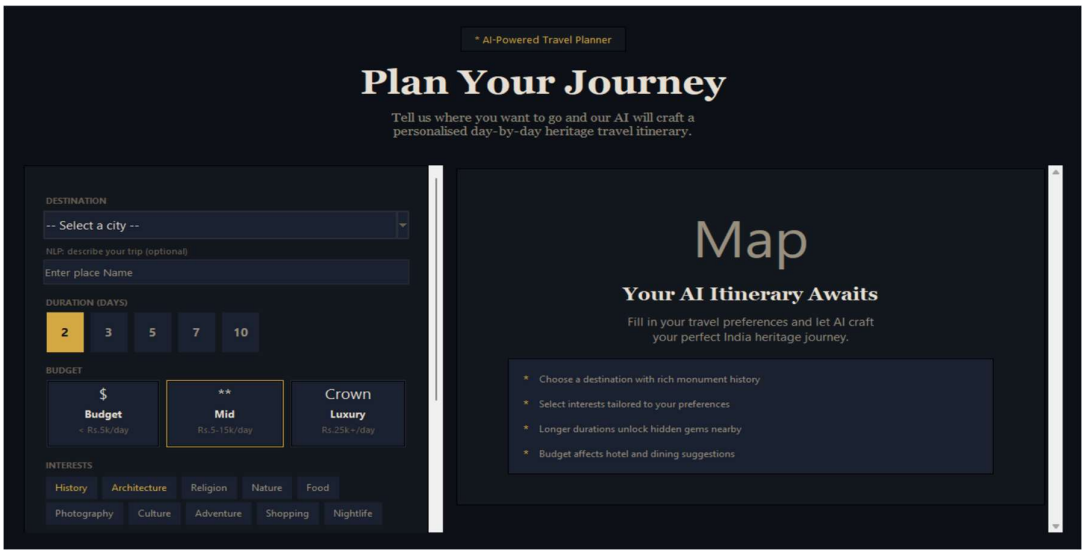
Project Abstract

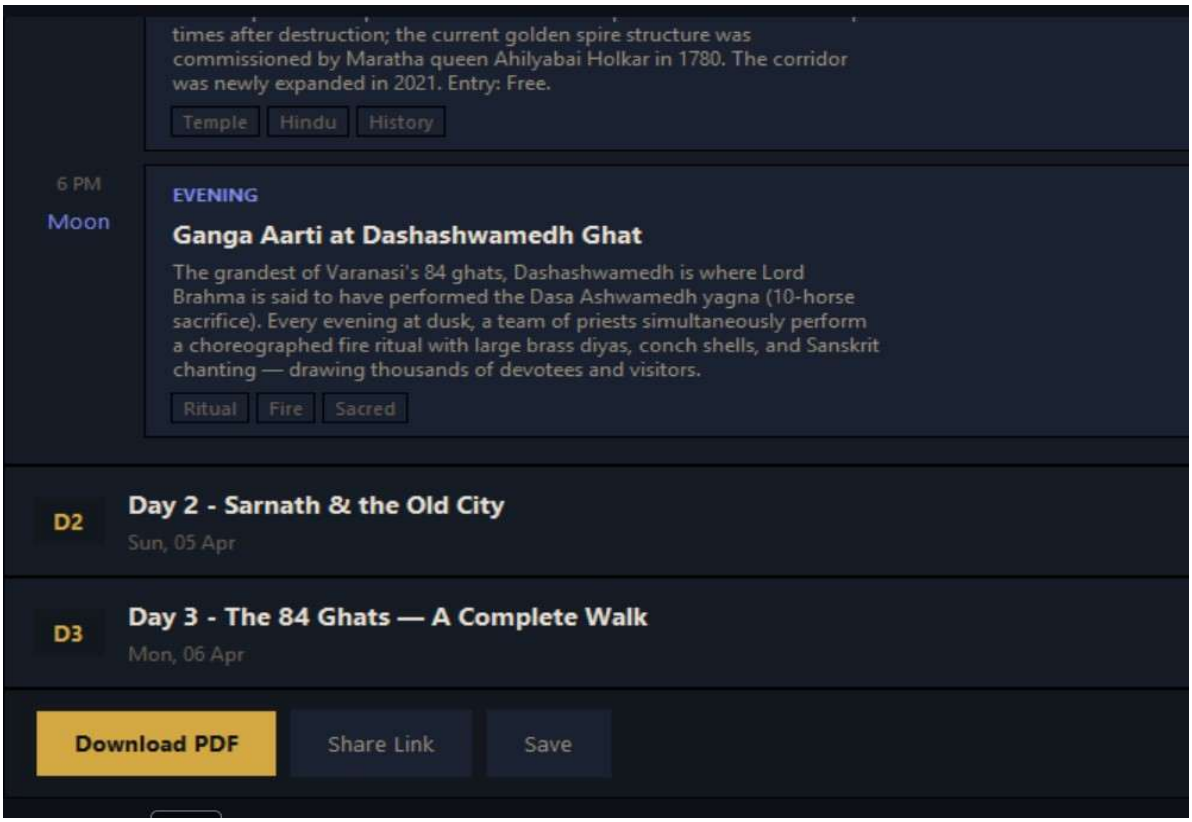
The **AI Travel Planner** is a sophisticated desktop application designed to simplify the complexity of travel itinerary creation. By leveraging a structured knowledge base of global heritage sites and cultural landmarks, the application generates comprehensive, time-blocked schedules. It provides users with a seamless transition from digital planning to a professional-grade physical or digital PDF document, ensuring all logistical details from morning excursions to evening relaxation are curated for an optimal travel experience.

Technical Specifications

- **User Interface:** A custom-themed Dark Mode interface built with Tkinter, featuring a gold-and-slate aesthetic for a premium user experience.
- **Document Generation:** Integration of the ReportLab library to programmatically generate A4-sized PDF itineraries with custom branding, table styles, and color-coded headers.
- **Concurrency Management:** Implementation of threading to handle intensive PDF generation and "AI simulation" steps (like route optimization and heritage mapping) without interrupting the UI responsiveness.
- **Data Architecture:** A multi-layered dictionary system that stores historical context, activity descriptions, and time-specific tags for over a dozen major cultural hubs.
- **Intelligent Recommendation Engine (KNN):** Implementation of a **K-Nearest Neighbours (KNN)** algorithm using scikit-learn. The system performs multi-dimensional vector matching between user preferences and a city-interest matrix. It utilizes **MinMaxScaler** for feature normalization and **Cosine Similarity** to ensure high-precision destination matching based on unique cultural and travel attributes.
- **Natural Language Processing (NLP) Module:** A custom-built, rule-based **NLP Engine** that utilizes regular expressions (re) and lexicon mapping. The engine

performs entity extraction to identify city names and intent classification to map user-typed keywords.





Key Features

- **Intelligent Itinerary Logic:** Users can define their destination, trip duration (up to 10 days), and travel pace. The system dynamically scales the content to fit the requested timeframe.
- **Budget Tiering:** Integrated estimation for **Budget**, **Mid-range**, and **Luxury** travel styles, helping users plan according to financial preferences.
- **Time-Block Scheduling:** Each day is divided into three distinct phases:
    - **Morning (7:00 AM):** High-energy heritage visits.
    - **Afternoon (1:00 PM):** Cultural immersion and museums. ○ **Evening (6:00 PM):** Local cuisine and leisure.
- **Professional Export Options:** * **PDF Report:** A formal document with embedded metadata and structured tables.
    - **Digital Share:** Generation of a temporary HTML/browser-viewable version for quick mobile access.

Walk Through History
Varanasi, Uttar Pradesh Heritage Journey
The Eternal City on the Ganges
3 Days
Mid Budget
Relaxed Pace
3
9
6
Rs.7,500
DAYS
EXPERIENCES
MONUMENTS
EST./DAY
D1
Day 1 - The Ghats & Sacred Fire
Saturday, 04 Apr 2026
7 AM
MORNING - 7 AM
Sunrise Boat Ride on the Ganges
Varanasi's 84 ghats stretch for 6.5km along the western bank of the Ganges. A sunrise boat ride (cost ~Rs.300–500) reveals the full spectacle of daily Hindu life — priests, pilgrims bathing, yoga practitioners, and the burning pyres of Manikarnika Ghat. This is considered one of the world's oldest continuously inhabited cities.
Sacred | Photography | Culture
1 PM
AFTERNOON - 1 PM
Kashi Vishwanath Temple
One of the 12 Jyotirlinges (sacred abodes of Lord Shiva) and among the most important temples in Hinduism. The temple has been rebuilt multiple times after destruction; the current golden spire structure was commissioned by Maratha queen Ahilyabai Holkar in 1780. The corridor was newly expanded in 2021. Entry: Free.
Temple | Hindu | History
6 PM
EVENING - 5 PM
Ganga Aarti at Dashashwamedh Ghat
The grandest of Varanasi's 84 ghats, Dashashwamedh is where Lord Brahma is said to have performed the Dasa Ashwamedh yagna (10 horse sacrifice). Every evening at dusk, a team of priests simultaneously perform a choreographed fire ritual with large brass diyas, conch shells, and Sanskrit chanting — drawing thousands of devotees and visitors.
Ritual | Fire | Sacred
D2
Day 2 - Sarnath & the Old City
Sunday, 05 Apr 2026
7 AM
MORNING - 7 AM
Sarnath — The Deer Park
Located 10km north of Varanasi, Sarnath is one of the four most sacred Buddhist pilgrimage sites. It was here, in the Isipatana Deer Park, that the Buddha delivered his first sermon (the Dhammacakkappavattana Sutta) in 5th century BCE. The Dhamek Stupa (500 CE) marks the exact spot. Entry: Rs.25 (Indians).
Buddhist | UNESCO | History

1 PM
AFTERNOON - 1 PM
Ramnagar Fort & Museum
Built in the 18th century by Maharaja Balwant Singh, this cream-coloured sandstone fort across the Ganges is the ancestral seat of the Maharaja of Varanasi. The Veda Vyasa Museum inside houses a remarkable collection of medieval weapons, royal palanquins, vintage automobiles, and rare astronomical instruments.
Fort | Royalty | Museum
6 PM
EVENING - 6 PM
Banarasi Silk Weaving Quarter
Varanasi's silk weaving tradition dates back over 2,000 years. The city's weavers — concentrated in the Peeli Kothi and Madanpura neighbourhoods — produce the celebrated Banarasi silk saris, known for their gold and silver brocade (zari) work. A guided workshop visit reveals the intricate handloom process. Cost: Free–Rs.200.
Craft | Culture | Shopping
D3
Day 3 - The 84 Ghats — A Complete Walk
Monday, 06 Apr 2026
7 AM
MORNING - 7 AM
Manikarnika Ghat — The Burning Ghat
Manikarnika is one of the most sacred cremation sites in Hinduism — Hindus believe dying here guarantees moksha (liberation from the cycle of rebirth). Fires have burned here continuously for 3,500 years according to tradition. The ghat is managed by the Dom community, hereditary keepers of the cremation fires. Approach with deep respect and no cameras. Entry: Free.
Sacred | Hindu | Culture
1 PM
AFTERNOON - 1 PM
Assi to Raj Ghat — Full Ghat Walk
Walk the 6.5km stretch of all 84 ghats from Assi Ghat (south) to Raj Ghat (north) — an extraordinary 3-hour journey through every dimension of Varanasi life. Each ghat has a distinct character: Harishchandra (a second burning ghat), Kedar Ghat (South Indian pilgrims), Darbhanga Ghat (a palace), and Pancheganga Ghat (five sacred rivers).
Walking | Culture | Heritage
6 PM
EVENING - 6 PM
Evening Aarti at Multiple Ghats
Beyond Dashashwamedh, smaller aartis happen simultaneously at Assi Ghat (more intimate, popular with locals), Rajendra Prasad Ghat, and Pancheganga Ghat. Witnessing the same ancient ritual at three different locations on the same evening gives a sense of the city's layered spiritual geography. Cost: Free.
Ritual | Sacred | Photography
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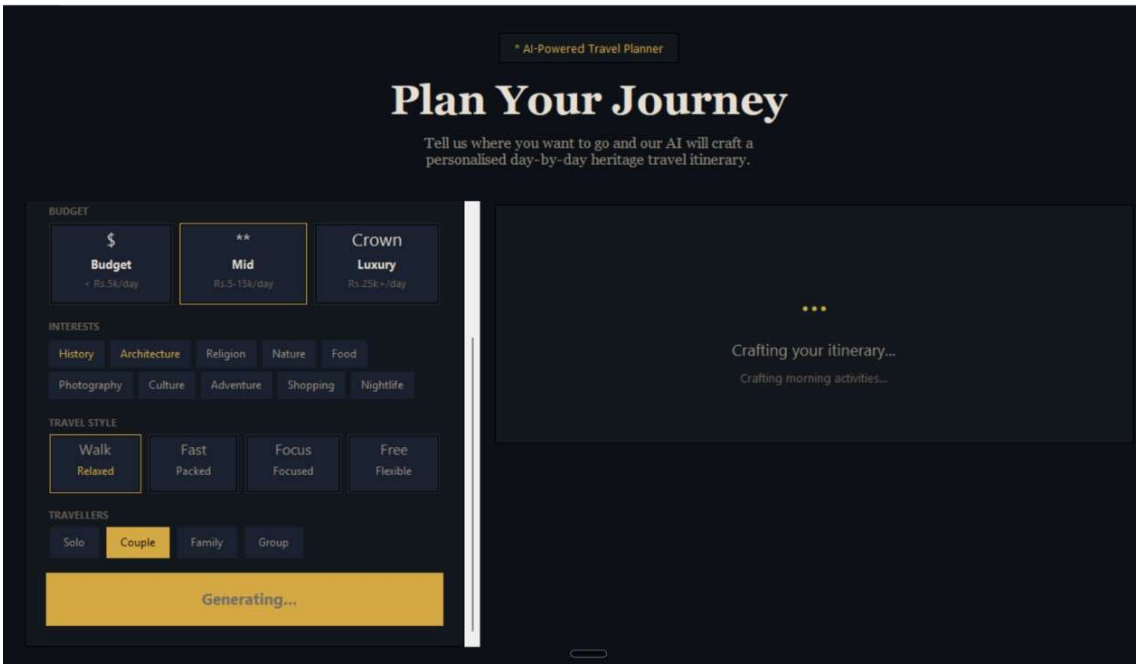
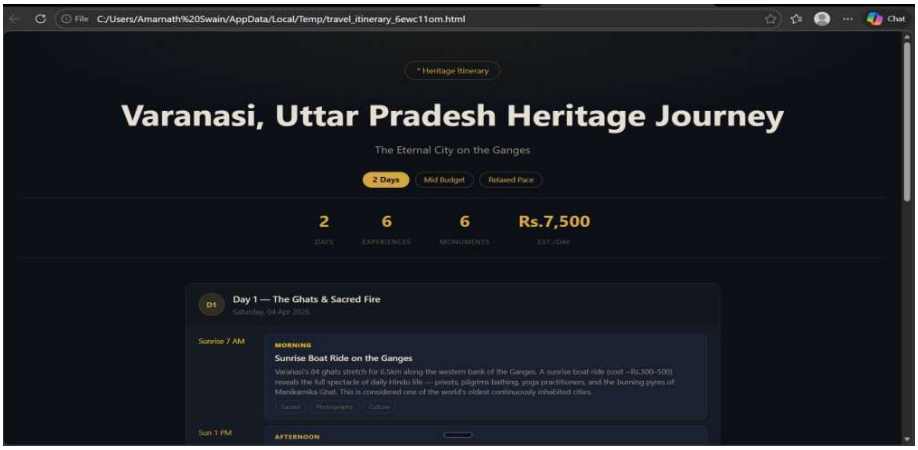

* AI-Powered Travel Planner
Plan Your Journey
Tell us where you want to go and our AI will craft a personalised day-by-day heritage travel itinerary.
BUDGET
$
Budget
< Rs.5k/day
**
Mid
Rs.5-15k/day
Crown
Luxury
Rs.25k+/day
INTERESTS
History
Architecture
Religion
Nature
Food
Photography
Culture
Adventure
Shopping
Nightlife
TRAVEL STYLE
Walk
Relaxed
Fast
Packed
Focus
Focused
Free
Flexible
TRAVELLERS
Solo
Couple
Family
Group
Generating...
Crafting your itinerary...
Crafting morning activities...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Chat
* Heritage Itinerary
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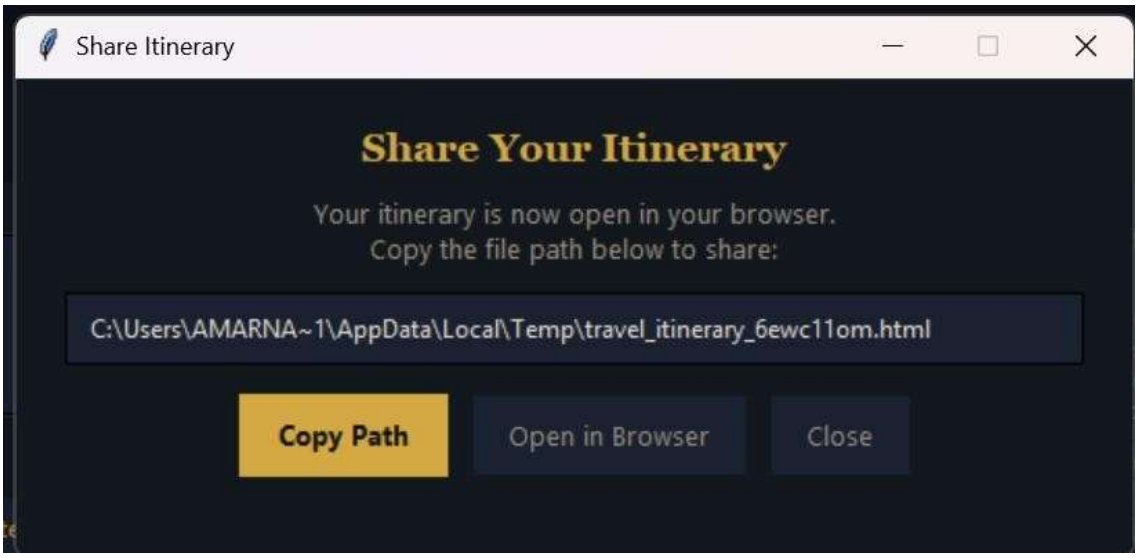
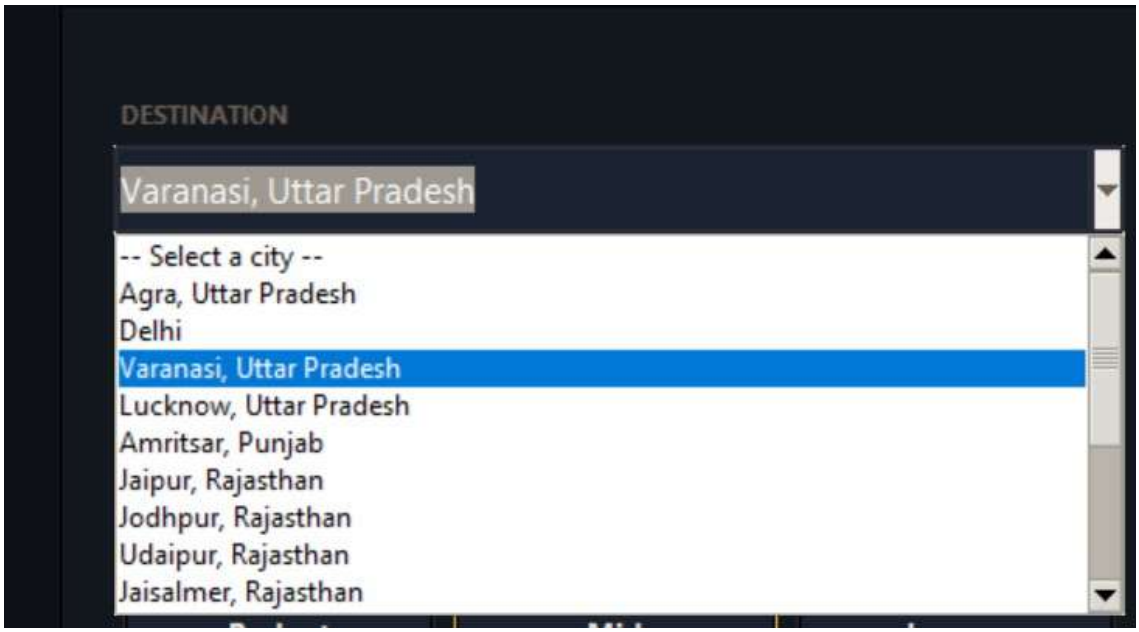
Sacred
Photography
Culture
Sun 1 PM
AFTERNOON



System Workflow

1. **Input Phase:** The user selects a destination (e.g., Agra, Delhi, Jaipur) and configures trip parameters via the dashboard.
2. **Simulation Phase:** The "AI Engine" performs a series of mock-processing steps, including mapping heritage sites by era and optimizing travel distances.
3. **Assembly Phase:** The SimpleDocTemplate engine gathers the data, applies Paragraph Styles, and constructs the visual layout.
4. **Output Phase:** The final report is saved to the local system and can be instantly opened for review or printing.



Future Enhancements

- **Real-time API Integration:** Connecting to live weather and Google Maps APIs for real-time travel alerts.
- **Machine Learning Integration:** Analysing user feedback to refine destination suggestions based on past preferences.
- **Multi-Language Support:** Localizing the generated reports for international travellers.

Conclusion

The AI Travel Planner successfully bridges the gap between automated data processing and usercentric design, providing a high-utility tool for modern travellers seeking structured exploration.

Thank You